Title: "Customer Purchase Propensity - Data Cleaning and Feature Engineering Pipeline"

Project Overview:

You are working as a **Junior Data Analyst** at an e-commerce company.
Your manager provides you with **multiple raw data sources** — including CSV, JSON, SQL database tables, and an open API — that contain customer and purchase-related information.

Your task is to **clean, preprocess, and engineer meaningful features** from the data to prepare it for a future **Machine Learning model** that predicts *whether a customer will make a purchase or not*.

You don't need to train a model — focus entirely on the **Data Preprocessing and Feature Engineering pipeline**.

🎯 Learning Objectives

By completing this project, students will demonstrate:

- Understanding and planning of a full data preprocessing workflow.
- Hands-on practice with data cleaning, imputation, encoding, outlier handling, transformation, and feature scaling.
- Awareness of how to handle different variable types for ML readiness.

🧩 Project Requirements

Step 1: Project Planning & Problem Framing

1. Briefly write a short plan (2–3 bullet points) explaining:
   - What is Data Analysis?
   - Steps in a Data Science project.
   - How to frame this dataset as a Machine Learning problem (binary classification).
   (Theory + short notes section)

Step 2: Data Import and Understanding

You are given:

- `customers.csv` – customer demographics and IDs
- `transactions.json` – transaction records
- `products.sql` – product info table (import using SQLite)
- API: `https://dummyjson.com/users` – for additional user details

Note: You can use the above 3 files for reference. You should populate with more data from the

Note: You can use the above 3 files for reference. You should populate with more data from the AI tool into that file, and then you can start your work.

Perform:

- Loading all datasets (CSV, JSON, SQL, API).
- Merge them using relevant keys (`customer_id`, `product_id`).
- Explore the combined dataset (use `.info()`, `.describe()`, `.head()`).

Step 3: Exploratory Data Analysis (EDA)

Perform EDA in three stages:

Univariate Analysis

- Visualize the distribution of numerical columns using histograms.
- Identify skewed variables.
- Use Pandas Profiling for automated insights.

Bivariate Analysis

- Correlate numerical features with the target column (`purchased` = 1 or 0).
- Compare means and visualize relationships (e.g., Income vs Purchase).

Multivariate Analysis

- Use pairplots, heatmaps, or grouped statistics to explore multiple feature relationships.

Step 4: Handling Missing Data

Use different imputation techniques:

- **Simple Imputer:** for numerical and categorical missing values.
- **Most Frequent Imputation:** for mode-based filling.
- **Missing Indicator + Random Sample Imputation:** show how missing flags are used.
- **KNN Imputer:** for multivariate imputation.
- **MICE Algorithm:** demonstrate chained imputation for correlated features.
- **Complete Case Analysis:** drop missing rows as a comparison.

Step 5: Outlier Detection & Handling

Apply at least 3 different techniques:

- **Z-score Method**
- **IQR Method**
- **Percentile Method**
- Optionally apply **Winsorization** to cap extreme values.

Step 6: Handling Mixed & Date/Time Variables

- Convert `signup_date` and `last_purchase_date` to datetime.
- Derive new features such as `days_since_last_purchase`.
- Handle mixed variables (like customer IDs embedded with letters).

Step 7: Encoding Categorical Data

Apply multiple encoding techniques:

- **Label Encoding**
- **One Hot Encoding**
- **Ordinal Encoding** (for education or satisfaction levels)

Also show:

- **Encoding Numerical Features** (e.g., binning income groups).

Step 8: Feature Scaling

Apply multiple scaling techniques to numeric data:

- **StandardScaler**
- **MinMaxScaler**
- **MaxAbsScaler**
- **RobustScaler**
- **Normalizer**

Use the `ColumnTransformer` to apply different transformations simultaneously.

Step 9: Feature Construction & Transformation

- Create new interaction features (e.g., `purchase_per_day = total_purchases / days_since_signup`).
- Apply transformations:
  - **FunctionTransformer** (Log, Square Root, Reciprocal)
  - **PowerTransformer** (Box-Cox and Yeo-Johnson)

Perform **binning** and **binarization**:

- Equal-width (quantile) binning on `income`.
- Binary conversion of purchase frequency (`frequent_buyer` = 1 if > threshold).

Step 10: Final Output

- Export the cleaned, transformed, and feature-engineered dataset to `processed_customer_data.csv`.
- Save preprocessing code as `feature_pipeline.py` or `.ipynb`.
- Submit a short summary:
  - What techniques were used?
  - What were the biggest issues with the raw data?
  - Which imputation and scaling worked best?

Deliverables

1. **Python Notebook** (`DataPreprocessing.ipynb`)
2. **Final CSV Output** (`processed_customer_data.csv`)
3. **Summary Report** (Theory + Observations, max 1 page)
4. All of the above three must be within a GitHub repository, including a Readme.md file.